



Power Quality Disturbance Classification using Machine Learning

Presented by_

Shrimathi G - 2127220601070

Sanjay Kumar V - 2127220601063



1. **Project Area** : Power Systems & Machine Learning
2. **Project Title** : Power Quality Disturbance Classification using Machine Learning
3. **Place of Work** : In-house
4. **Supervisor Name** : Mr. N. Suresh
5. **Project Group** : Batch 15

Motivation

- There is a clear gap between high-accuracy research models and practical, deployable solutions.
- **Industries need low-latency, edge-compatible intelligence**, with increasing renewable integration, **grids are becoming more unstable**, making real-time PQ monitoring essential.
- **Traditional techniques (FFT, RMS)** fail to capture complex, transient, and non-linear disturbances
- A scalable solution must **balance accuracy, speed, and computational efficiency** - current systems don't.
- This project targets a practical shift, from delayed detection to real-time intelligent classification at the edge.

Problems Identified and Solutions Suggested

Problems Identified

- Frequent PQ Disturbances: Over **60%** of grid issues stem from voltage sags, swells & harmonics causing **15–20% energy loss** and **30% shorter equipment life**.
- Outdated Detection: Conventional FFT/RMS systems **miss ~40% of transients** and analyze events **5–10 seconds late**, risking downtime.
- High Computational Demand: Existing AI models need **>2 GB GPU RAM** and cloud connectivity, adding **300–400 ms latency** impractical for field deployment.

Solutions Suggested

- Accuracy: Classifies PQ disturbances with **>95% accuracy** and **<100 ms inference time**.
- Edge AI Implementation: Cutting latency by **80%** and power cost by **40%**.
- IoT-Based Monitoring: Sensors stream **10 kHz data** to a live Node-RED dashboard, improving response **speed by ~70%**.

Literature Survey

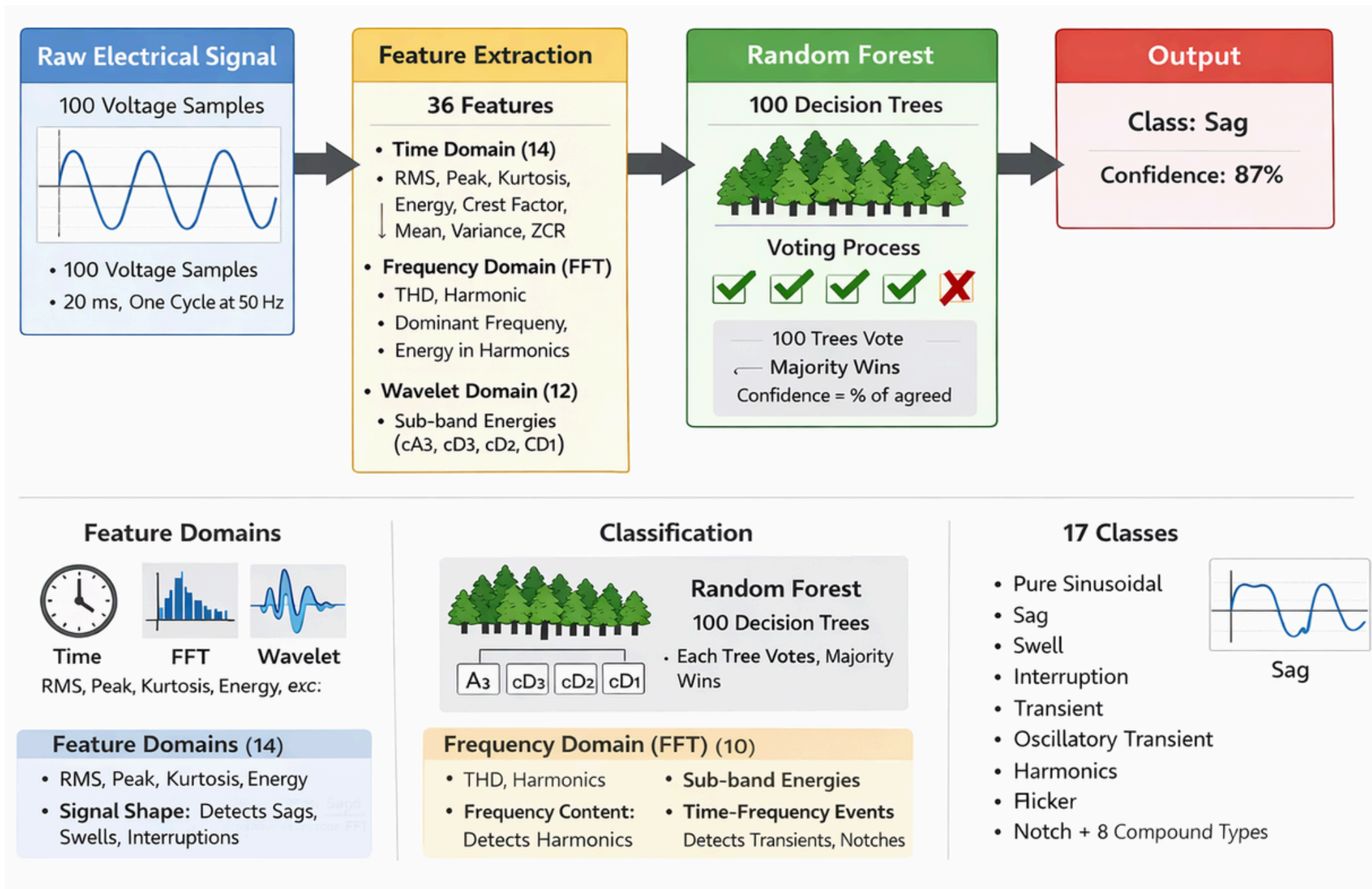
PAPER TITLE & AUTHOR	WORK	OUTCOME
Power Quality Disturbance Classification using Random Forest with Feature Selection for Real-Time Monitoring – Rahul, A. Sambher & P. Sinha, Lecture Notes in Electrical Engineering, Springer, 2025	Applied Random Forest classifier on PQD dataset with RMS, THD, crest factor, and spectral band features; studied feature count vs. accuracy trade-off for real-time deployment.	Achieved 97.50% accuracy — confirms RF is effective for real-time PQ classification but tested only on single disturbances without compound classes like Sag+Harmonics.
Power Quality Disturbances Categorization using Identity Feature Vector and Extreme Learning Machine – H. Rezapour et al., Results in Engineering, Elsevier, 2024	Generated 12 PQD classes including Sag, Swell, Harmonics, Flicker, Sag+Harmonics, Sag+Flicker, and Swell+Harmonics using wavelet-based identity feature vectors classified by ELM.	10 out of 12 classes achieved 100% accuracy; Swell+Harmonics was the hardest at 98% — our work addresses this using Random Forest with stronger spectral features.
Detection and Classification of PQDs using SPWVD and Vision Transformer – K. Ozbay & M. Gol, AIMS Energy, vol. 13, no. 5, 2025	Simulated 11 PQD types per IEEE 1159 in MATLAB (sag, swell, interruption, harmonics, flicker, harmonics with sag/swell, flicker with sag/swell) and classified with ViT model.	99.86% overall accuracy but requires image transformation and heavy GPU — our RF approach works directly on extracted features with no image conversion.

Literature Survey

PAPER TITLE & AUTHOR	WORK	OUTCOME
A Comprehensive Research of Machine Learning Algorithms for PQD Classifier Based on Time-Series Window – S. S. Tezcan et al., Electrical Engineering, Springer, 2025	Benchmarked RF, XGBoost, LightGBM, and kNN on a 21-class PQD dataset including compound disturbances at SNR 20–50 dB per IEEE 1159 and IEC 61000 standards.	RF delivered competitive accuracy — validates our choice of RF as classifier for a 12-class compound PQD problem.
Perception of PQDs using Fourier, STFT, CWT and DWT – S. Lather et al., Scientific Reports, Nature, Feb. 2024	Mathematically modelled Sag, Swell, Interruption, Harmonics, Flicker in MATLAB using parametric equations per IEEE 1159; analysed signals using FFT, STFT, CWT, and DWT transforms.	Validated MATLAB signal generation equations for all PQD types used in our project; confirmed FFT-based features (THD, harmonic magnitudes) are most discriminative.



System Design



Training Phase (done once):

- Load XPQRS dataset (17,000 signals, 17 classes, 100 samples each)
- Extract 36 features from each signal (time + FFT + wavelet)
- Split 80% train / 20% test
- Normalize features using StandardScaler
- Train Random Forest classifier (100 decision trees)
- Validate with 5-fold cross-validation
- Save trained model to disk (.pkl)

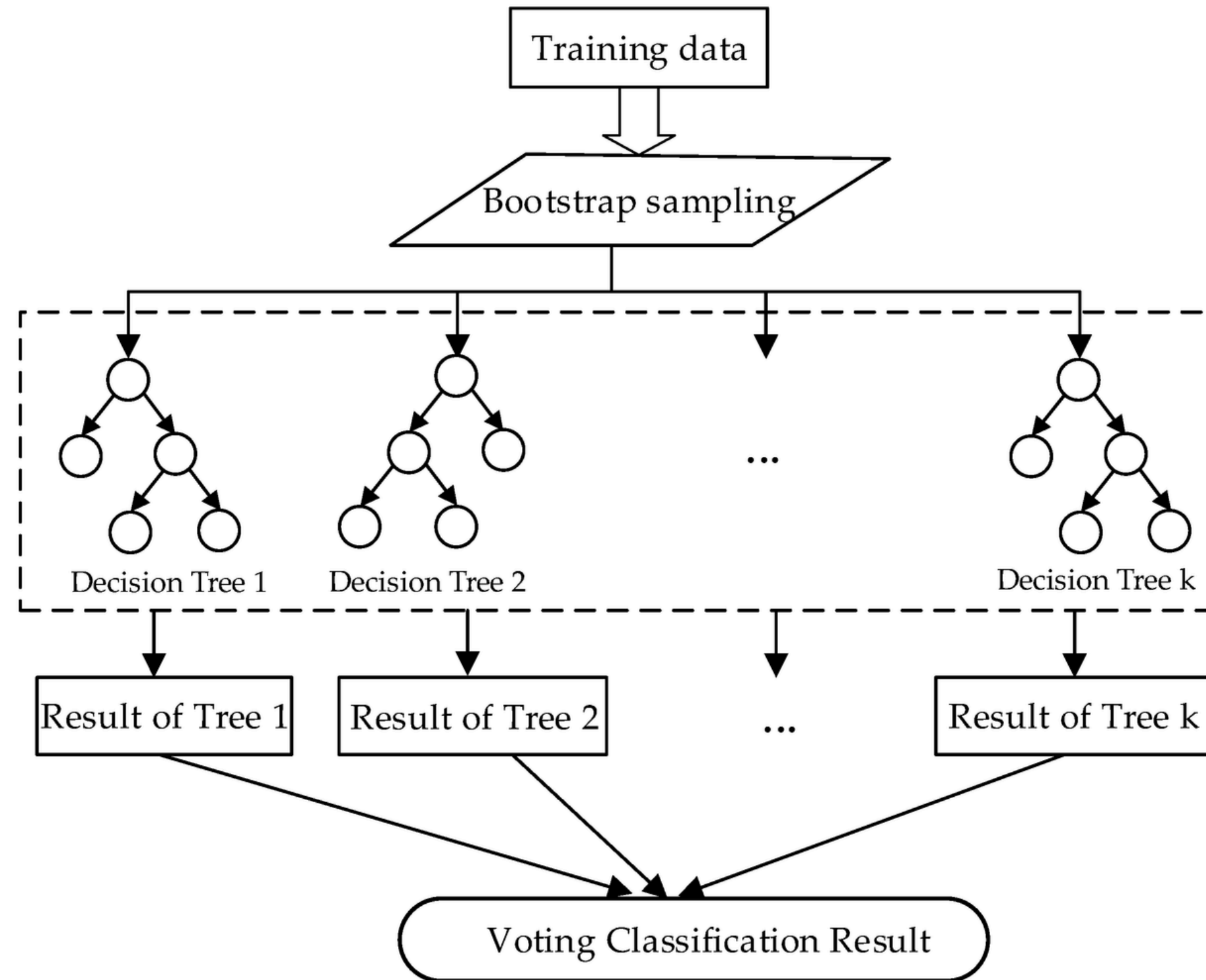
Prediction Phase (runs anytime):

- Take new signal (100 voltage values)
- Extract 36 features
- Load saved model and predict
- Output: Normal/Abnormal + disturbance type + confidence score

Objectives of the Project

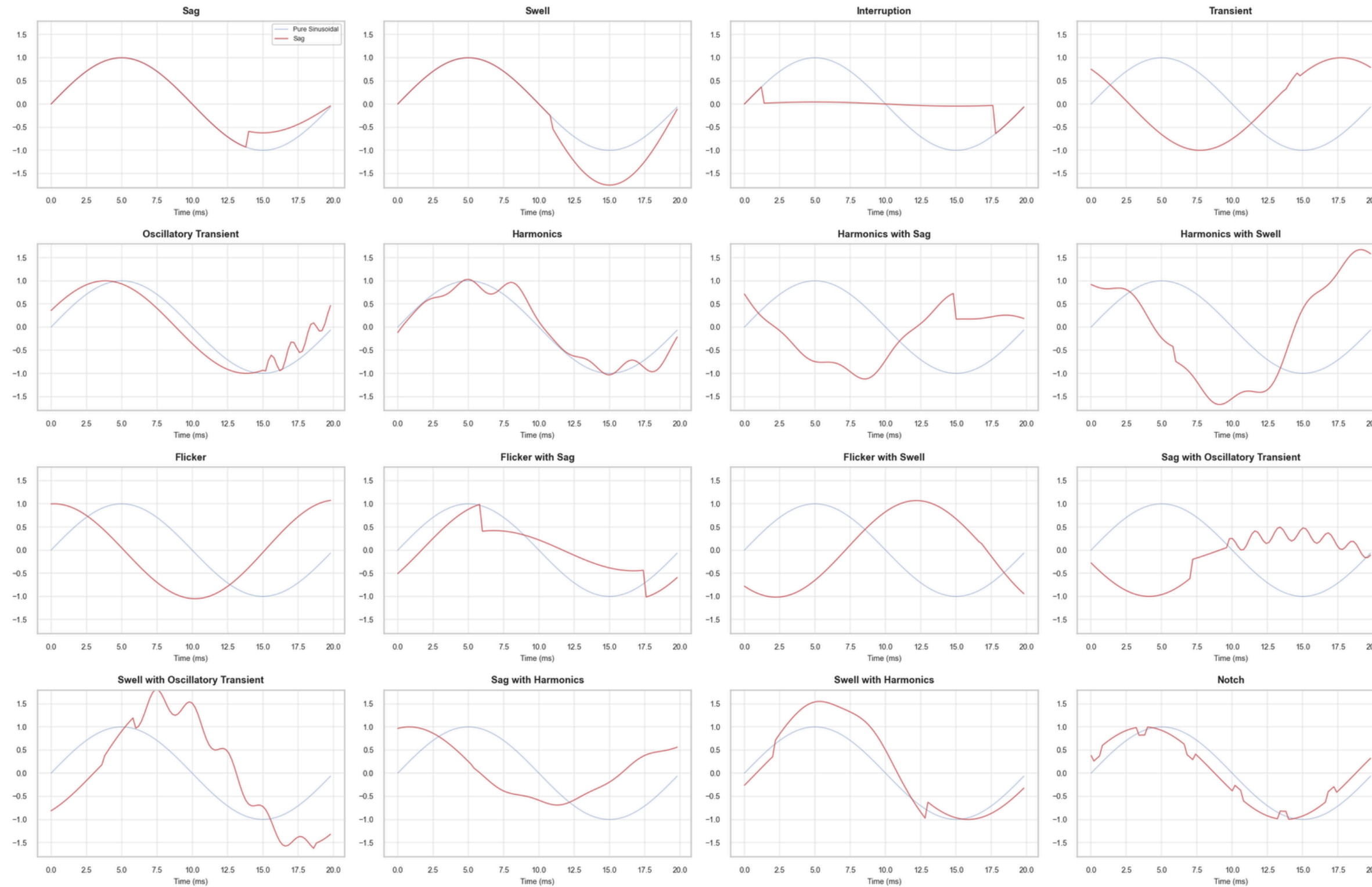
- Develop a robust machine learning model to accurately **classify multiple types of power quality disturbances** under real-world conditions.
- **Achieve high classification accuracy (>95%) with low inference time (<100 ms)** for practical deployment.
- **Enable edge-compatible implementation**, reducing dependency on high-end GPUs and cloud infrastructure.
- Extract and **optimize multi-domain features (time, frequency, wavelet)** to improve model reliability across disturbance types.
- Provide clear output including disturbance type, severity indication, and confidence score for actionable insights.

Flowchart

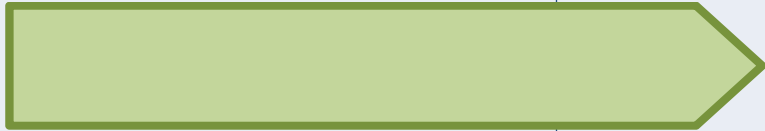
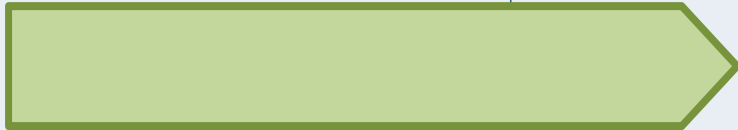


Disturbances

Disturbance vs. Pure Sinusoidal Reference



Plan of Action

	DECEMBER 2025	JANUARY 2026	FEBRUARY 2026	MARCH 2026
TASK				
Literature Survey				
Design				
Simulation				
Simulation and Software Integration				
Report Preparation				

SDG

SDG Goals	Project Contribution
SDG 7 – Affordable and Clean Energy	Improves energy efficiency and grid stability by enabling real-time detection of power quality disturbances, ensuring reliable and sustainable electricity supply.

References

- [1] Rahul, A. Sambher, and P. Sinha, "Enhancing real-time power quality monitoring: a comprehensive study on feature selection and performance analysis of random forest classifiers," Lecture Notes in Electrical Engineering, vol. 1305, Springer, Singapore, 2025.
- [2] S. Janthong and P. Phukpattaranont, "Power quality disturbance detection using improved grasshopper optimization and adaptive boosted random forest," Electrical Engineering, Springer, 2025.
- [3] S. Wei, D. Wenjuan, and C. Xia, "Power quality disturbances categorization using identity feature vector and extreme learning machine," Intelligent Systems with Applications, Elsevier, vol. 24, p. 200446, 2024.
- [4] P. M. Peruman and K. Ayyar, "Power quality disturbance identification using hybrid deep learning in renewable energy systems," Scientific Reports, Nature, vol. 15, 2025.
- [5] S. S. Tezcan, U. Yildirim, and H. Dogan, "A comprehensive research of machine learning algorithms for power quality disturbances classifier based on time-series window," Electrical Engineering, Springer, vol. 106, pp. 3983-4001, 2024.
- [6] R. J. J. Molu, W. F. Mbasso, K. T. Saatong, and S. R. D. Naoussi, "Enhancing power quality monitoring with discrete wavelet transform and extreme learning machine: a dual-stage pattern recognition approach," Frontiers in Energy Research, vol. 12, p. 1435704, 2024.
- [7] S. Jain, A. Satsangi, R. Kumar, and D. Panwar, "Intelligent assessment of power quality disturbances: a comprehensive review on machine learning and deep learning solutions," Computers and Electrical Engineering, vol. 123, Elsevier, 2025.
- [8] M. S. Priyadarshini, M. Bajaj, L. Prokop, and M. Berhanu, "Perception of power quality disturbances using Fourier, short-time Fourier, continuous and discrete wavelet transforms," Scientific Reports, Nature, vol. 14, p. 3443, Feb. 2024.
- [9] M. S. Priyadarshini, M. Bajaj, and I. Zaitsev, "Energy feature extraction and visualization of voltage sags using wavelet packet analysis for enhanced power quality monitoring," Scientific Reports, Nature, vol. 15, p. 2226, Jan. 2025.